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MAY/JUNE 2007

FORM TP 2007197

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

25 MAY 2007 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.

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- 425-4494.

2 - Cultivation of marine org. in seawater, saltwater ponds or raceways. marine org → kelp seamoss.
3 - Org. being cultivated feed on naturally occuring algae and plankton.
4. Exclusive use is made of salt water in the process.
5 Depending on the species being cultivated, mariculture may be practiced in brackish water
6. In systems where ponds are utilized nutrients are provided.

Ad- Alternative source of protein
1- commercial fishing has ~~reduced~~ sulted in over fishing and depletion of natural fish stocks.
2- This has increased cost related to the provision of - 2 - fish protein or alternative sources of proteins.
3- Mariculture could provide/produce significant quantities of additional protein (food) for growing Caribbean populations at a reasonable cost.
Ad- ~~cos~~ Lower cost per unit yield.
1- Mariculture requires little input of food and energy

MODULE 1

Answer ANY TWO questions.

1. (a) Outline FOUR features of mariculture. **[4 marks]**

(b) "Mariculture has tremendous potential to contribute to the economies of Caribbean countries."

Evaluate the statement above. Include TWO advantages and TWO disadvantages of mariculture in your response. **[16 marks]**

not phrase properly

Total 20 marks

reparian buffer zone. organic farming. Agroforesty. Intercropping, mix, crop rotation. mulching. Shifting Cultivation.

2. (a) (i) State TWO types of sustainable agricultural practices. **[2 marks]**

(ii) Describe BOTH of the practices stated in 2 (a) (i). **[6 marks]**

(iii) State TWO advantages of EACH sustainable agricultural practice stated in 2 (a) (i). **[4 marks]**

(b) Explain why commercial agricultural systems require high inputs of inorganic fertilizers to maintain yields. **[4 marks]**

(c) Outline ONE environmental impact resulting from the use of large amounts of inorganic fertilizers in commercial agricultural systems. **[4 marks]**

Total 20 marks

fallowing. soil compacting

3. Figure 1 shows the yields obtained from rice fields by farmers in Bangladesh with both normal and reduced pesticide use between the period 1970 to 1995.

(a) Describe the trends illustrated by EACH line graph in Figure 1. (Include in your description at least 3 points for each line graph.) **[6 marks]**

(b) Describe TWO alternative methods that the farmers in Bangladesh may adopt for controlling pests. **[6 marks]**

(c) Outline ONE advantage and ONE disadvantage of the use of EACH of the alternative methods for controlling pests described in 3 (b). **[8 marks]**

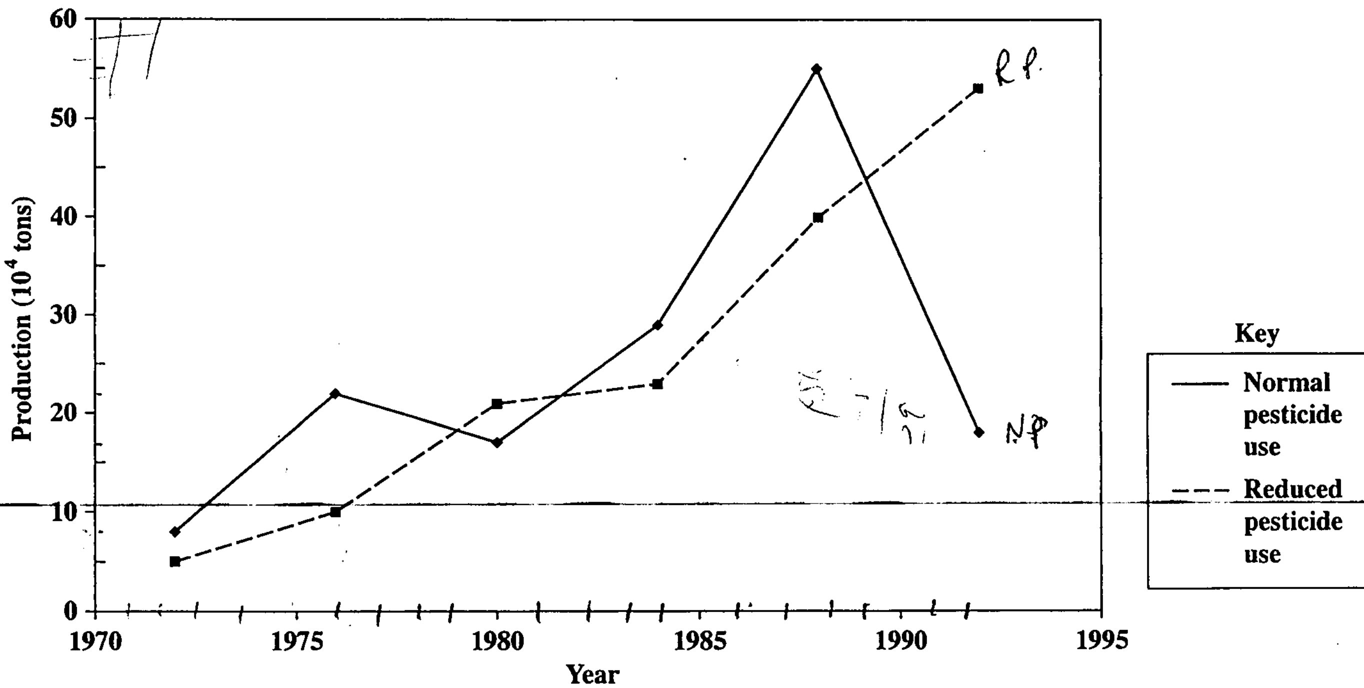


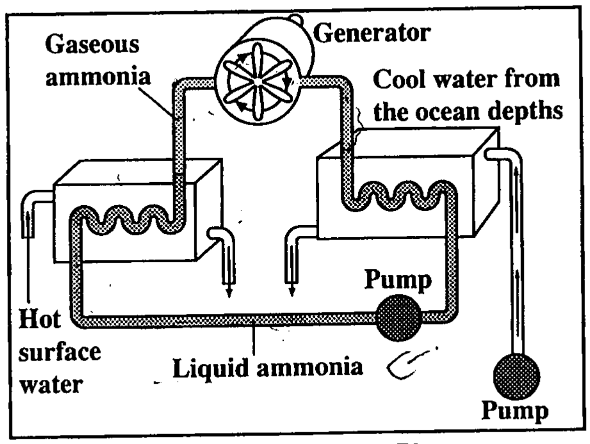
Figure 1. Rice field production with pesticide use

Total 20 marks

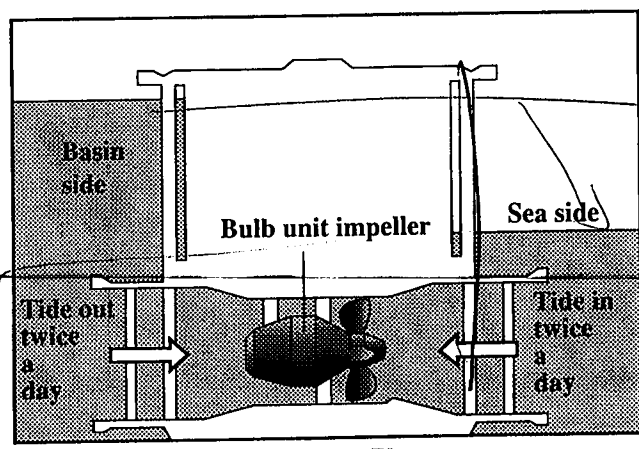
MODULE 2

Answer ANY TWO questions.

4. Figure 2 shows the essential features of an ocean thermal power plant and a tidal power plant.



Ocean Thermal Power Plant



Tidal Power Plant

Figure 2. Ocean thermal and tidal energy systems

(a) Distinguish between 'renewable energy source' and 'non-renewable energy source'. **[2 marks]**

(b) With reference to Figure 2, describe the basic principle of how energy is harnessed in EACH plant. **[8 marks]**

(c) Assess the feasibility of ANY ONE of the systems in Figure 2 to satisfy the energy needs of your country. In your assessment include at least TWO advantages and TWO disadvantages of the system. **[10 marks]**

Total 20 marks

5. (a) Outline TWO reasons why conventional power plants in Caribbean countries are located along coastlines. **[6 marks]**

(b) Discuss TWO environmental impacts on coastal environments resulting from the operation of power plants in coastal areas. **[6 marks]**

(c) Explain how cogeneration may be used to decrease the impact of power plants on coastal waters and improve their efficiency of operation. **[8 marks]**

efficiency 30-35 % → (50-60)

Total 20 marks

6. (a) Describe TWO environmental impacts resulting from the use of petrol as the energy source for motor vehicles. **[6 marks]**

(b) Discuss TWO approaches that Caribbean countries may adopt to decrease the environmental impacts from motor vehicles. 1- Catalytic converters 2- ~~Regular~~ inspection, Removing of defective vehicles. **[8 marks]**

(c) Evaluate the statement below.

"Cars powered by electricity will be the solution to the problems resulting from the use of petrol as the energy source in motor vehicles." **[6 marks]**

Total 20 marks

Tidal power plant
- problem siltation.

Check wind energy generates
a large amount of energy

warm.

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MODULE 3

Answer ANY TWO questions.

7. Figure 3 shows the concentration of hazardous chemicals in the soil of a particular region in Country Y over a ten-year period.

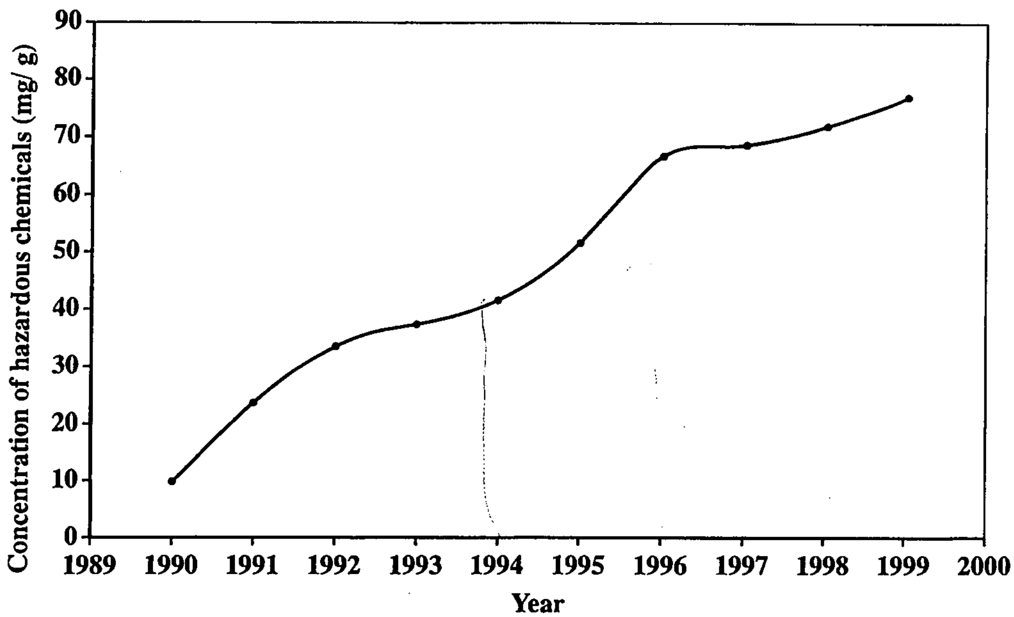

Figure 3. Concentration of hazardous chemicals in soil

~~(a) State the names of TWO hazardous chemicals that may be found in soils. [2 marks]~~

(b) Describe the trend indicated in Figure 3. **[2 marks]**

(c) Explain why the trend described in 7 (b) above poses a significant threat to humans and other organisms in Country Y. Include THREE reasons with appropriate examples in your response. Bioaccumulation, Biomagnification, Synergism, Carcog **[9 marks]**

(d) Explain how human activities in surrounding countries contribute to the trend described in 7 (b). **[7 marks]**

Caribbean Countries close together.
Chemicals from Industrial processes from neighboring country may be emitted into the atmosphere

Total 20 marks

Explain. Cancer.

Selenium C used to extract iron ores

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8. The results of the sampling of water from River Z are recorded in Table 1, and selected water quality standards for drinking and recreation are presented in Table 2.

TABLE 1: WATER QUALITY PARAMETERS OF RIVER Z

Site	Number of readings	Nitrates(mg/l)	Chloride(mg/l)	Faecal Coliform/100 ml
1	9	0.36	200	2400
2	8	0.28	225	1600
3	9	0.48	230	2000
4	10	0.50	220	1800
5	8	0.54	210	2600

2599

TABLE 2: SELECTED WATER QUALITY STANDARDS

Drinking Water Quality Standards	
Nitrates	< 44 mg/l
Chloride	< 250 mg/l
Total dissolved solids	< 1000 mg/l
Recreational Water Quality	
Faecal coliform	< 2400/100 ml

(a) What is meant by the term 'water quality standard'? [2 marks]

(b) Using the data presented in Table 1 and Table 2, assess the suitability of using River Z

(i) for the purposes of recreation [3 marks]

(ii) as a source of drinking water. [3 marks]

(c) Outline TWO reasons for the presence of nitrates in the water. [6 marks]

(d) Describe TWO environmental impacts of nitrates on the river ecosystem. [6 marks]

Total 20 marks

(b) - Comparison of all sites - 1

any other appropriate analysis - 1 each

different sites - 1 mks

Eutro.

reduces sunlight penetration and photosynthetic ability of water dwelling plants.

(c) - Agriculture percolate or dissolve into ground water.

skip

flow into man-made gullies which

9. Figure 4 shows the structure of the atmosphere.

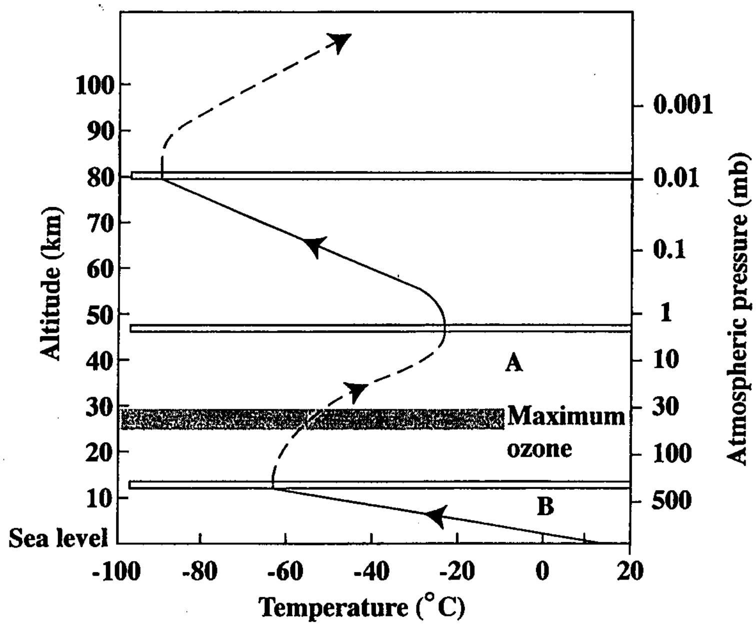

Figure 4. Structure of the atmosphere

(a) Identify the layers labelled A and B in Figure 4. **[2 marks]**

(b) State the name of TWO gases that play a predominant role in global warming. **[2 marks]**

(c) With reference to Figure 4, explain how global warming occurs. **[8 marks]**

(d) Assess the feasibility of TWO measures that have been proposed to slow global warming. **[8 marks]**

Total 20 marks

END OF TEST

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